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Advances and Future Prospects of Vision-Language-Action Models in Autonomous Driving

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***Abstract*—Autonomous driving research has transitioned from rule-based modular approaches to data-driven end-to-end methods. However, pure end-to-end systems often lack interpretability and generalization. With the rapid advancement of multimodal large models, knowledge-driven large model approaches are increasingly gaining attention from academia and industry. Vision-language-action (VLA) models, by integrating visual perception, linguistic reasoning, and action generation, provide a unified framework to enhance system interpretability and adaptability in complex environments. This paper systematically reviews the state-of-the-art of VLA models in autonomous driving. It first elucidates the necessity of VLA models in this domain. Then, it summarizes their typical architectures, inference processes, and associated datasets. Finally, it discusses current challenges and proposes future research directions.**

Keywords—autonomous driving, Vision-Language-Action Model, multimodal integration, interpretability, generalization, knowledge-driven

I. Introduction

Autonomous driving technology is rapidly evolving with advancements in deep learning, reinforcement learning, and hardware computational capabilities [1]. Research has shifted from traditional modular approaches to data-driven end-to-end methods. However, current end-to-end approaches, despite simplifying system architecture, suffer from poor interpretability [2], limited generalization [3], causal confusion [4], and data distribution shifts [5], leading to suboptimal or erroneous decision-making in complex scenarios.

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In recent years, vision-language models (VLMs) pretrained on internet-scale data have made significant progress and have been initially applied to autonomous driving. For instance, some studies [6][7] leverage VLMs to analyze vehicle camera inputs, providing natural language explanations of driving behaviors or answering queries about driving scenarios, thereby enhancing system interpretability. However, this paradigm typically employs VLMs as a high-level reasoning or explanation module, with outputs loosely coupled to downstream planning and control, failing to fully integrate the rich prior knowledge and reasoning capabilities of language with real-time vehicle action generation.

To address these challenges, researchers are exploring vision-language-action (VLA) architectures that unify environmental perception, logical reasoning, and action control [8]. Yet, a systematic review of VLA models' principles, applications, challenges, and future directions in autonomous driving remains absent. This paper provides a comprehensive survey, first outlining the evolution of autonomous driving, then highlighting the necessity of VLA models, followed by an overview of their typical architectures and inference processes, a summary of relevant datasets, and finally, an analysis of current challenges and future research directions.

II. The Development of Autonomous driving

The development pathways of autonomous driving can be categorized based on their core driving forces into rule-based modular approaches, data-driven end-to-end approaches, and knowledge-driven large model approaches, as shown in Fig. 1.

A. Rule-based Modular Approach

The rule-based modular approach decomposes complex driving tasks into independent subtasks, including perception, localization and mapping, decision planning, and control [9], as shown in Fig. 1(a). Typical modular systems achieve

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localization using high-definition maps and global navigation satellite systems, employ multi-sensor fusion with detection and recognition algorithms for perception, and utilize human-designed logical rules or optimization algorithms for decision planning, with controllers executing specific vehicle actions. This modular design offers strong interpretability and debuggability, facilitating independent development and validation [10]. However, its performance is constrained by the completeness of human-defined rules, and the cascaded architecture can lead to error accumulation from upstream modules, compromising the reliability of final decisions.

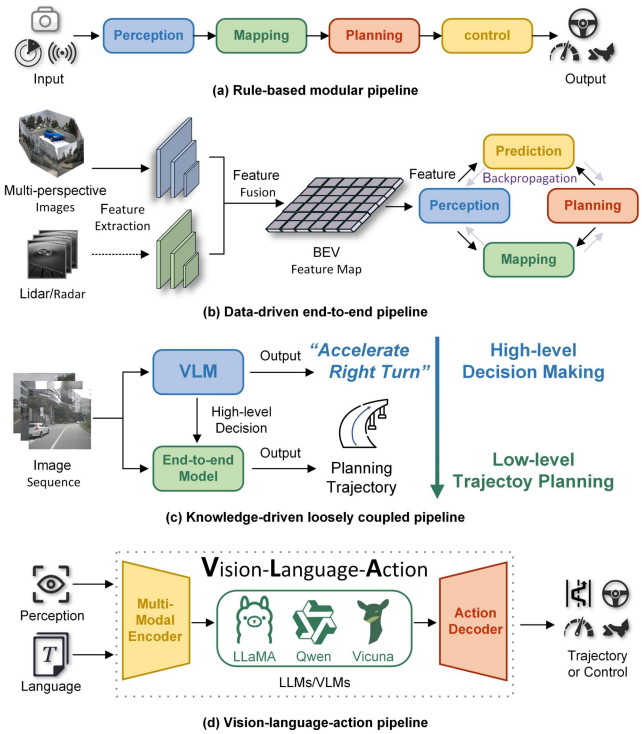

Fig. 1. The Development of Autonomous driving [18].

B. *Data-driven End-to-end Approach*

The data-driven end-to-end approach employs deep neural networks to directly map raw sensor inputs to decision outputs, such as trajectories or control signals. Early work, exemplified by NVIDIA's PilotNet [11], used convolutional neural networks to regress steering angles. Current mainstream methods typically adopt Transformer-based architectures [12]-[14], converting multi-sensor data into Bird's-Eye-View space for trajectory planning, as shown in Fig. 1(b). This approach minimizes explicit intermediate modules, reduces information loss, and learns human-like driving strategies from large-scale driving data. However, neural network-based models are often considered "black boxes," lacking transparency and trustworthiness in decision-making [15]. Moreover, their performance heavily depends on the scale and diversity of training data, with limited generalization to out-of-distribution scenarios.

C. *Knowledge-driven Large Language Model Approach*

The knowledge-driven large model approach leverages the world knowledge and general reasoning capabilities of pretrained large models to endow autonomous driving systems with human-like common-sense understanding and decision-making logic [16]. Previous studies [17]-[20] often adopted a loosely coupled hybrid architecture, where a language model serves as a high-level cognitive and reasoning module, deployed in parallel with an independent end-to-end model handling specific driving tasks, as shown in Fig. 1(c). This approach significantly enhances system interpretability and cognition in unconventional scenarios. However, the separation of reasoning and execution introduces a "loose coupling" issue, causing delays and information gaps between the language model's cognitive outputs and the vehicle's physical actions, hindering an efficient closed-loop driving system. To address this, the knowledge-driven approach is evolving toward a VLA paradigm that deeply integrates perception, reasoning, and action.

The VLA model concept, first proposed by Google in robotics [21], aims to enable embodied agents to generate physically feasible actions based on visual perception and language instructions. As a key application of embodied intelligence, autonomous driving is a critical domain for exploring VLA architectures [8]. In autonomous driving tasks, VLA models unify visual inputs, language instructions, or reasoning chains with action outputs within an autoregressive or diffusion-based generative model, transforming traditional perception, prediction, and planning processes into a language generation task for direct mapping and optimization of driving actions, as shown in Fig. 1(d).

III. The Architecture and Methodology of VLA

A. *Typical Architecture*

The VLA model integrates visual perception, language understanding, and driving strategy generation, and is mainly composed of three modules: a visual encoder, a language model, and an action decoder, as shown in Fig. 2.

1) Vision encoder: the vision encoder transforms visual inputs into structured feature vectors. It processes raw pixel data, such as single-frame images or multi-view video sequences, to produce semantic feature vectors, enabling cross-modal interaction with language models. Current mainstream visual encoders include BEV encoder [28][29], ViT [22] [25] [27] [30] [31] [33] [39] [51], CLIP [23] [24] [32] [35][36][50], and composite models [33][37].

2) Language model: the language model, pretrained on extensive internet text data, serves as a universal language understanding and generation module. As the core reasoning component, it processes feature vectors from the visual encoder, text instructions, and vehicle state, performing cross-modal alignment to generate natural language outputs, such as driving decisions, scene descriptions, control commands, or action tokens. Commonly used language models include LLaMA [23][27][28][33][35][36][37][50], its Vicuna variants

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[24][30][32][37][39], the Qwen series [21] [27] [29] [31] [33] [37][50][51], and the Gemini series [26].

3) Action decoder: the action decoder converts multimodal semantic representations into executable actions. In the VLA architecture, it typically processes instruction semantics from the language model to generate action signals. Action decoders primarily rely on autoregressive Transformer decoders [21] [23] [26] [27] [28] [29] [33] [34] [35] [36] [37] [38] [39] [51] or diffusion models [24] [25] [30] [31] to produce action sequences.

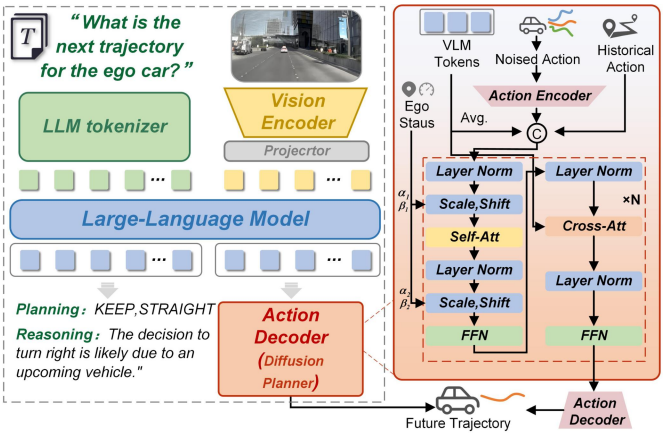

Fig. 2. The Typical Architecture of VLA [31].

B. Inference Process

The typical VLA inference process is as follows: The model receives multimodal data, such as sequences of camera images, textual information like language instructions, questions, and target point coordinates, as well as ego-vehicle status information such as speed. The textual information is usually embedded using the tokenizer that comes with the language model. Visual information is embedded as visual tokens through a visual encoder. To handle high-resolution images, they are typically divided into small patches and encoded independently, then aligned with language tokens through adaptation operations. The ego-vehicle status is aligned through a multilayer perceptron. After encoding, information from different modalities can be transformed into a unified embedding sequence for input into the language model through concatenation or cross-modal alignment operations such as modality fusion. The language model is based on a pre-trained multimodal large language model, which is usually fine-tuned with techniques such as LoRA [41] to adapt to autonomous driving tasks. The output of the language model includes both text and actions. The text is predicted through autoregressive token prediction and is used for task responses or explanations of driving decisions. Actions can be a sequence of trajectory points in a language-like form, or specially designed action or trajectory point tokens, which are then decoded into specific control signals or trajectories by an action decoder.

C. Dataset

VLA model datasets require rich sensor signals and natural language descriptions linked to visual content. NuScenes [42], Waymo Open Dataset [43] and NavSim [45] are large-scale multimodal datasets with diverse sensor data and detailed 3D object annotations, mainly used to enhance the model's perception capabilities. CARLA [44], an open-source autonomous driving simulator, generates varied virtual driving data, commonly used for training and testing end-to-end and reinforcement learning models. BDD-X [46] and BDD100K [47] are real-world driving video datasets covering diverse scenarios and weather conditions, offering rich perception annotations, with BDD-X including language explanations. Reason2Drive [48] with over 600,000 video-text pairs, supports interpretable reasoning in complex driving environments. DriveLM [49], designed for VLA, integrates visual inputs, driving behaviors, and corresponding natural language explanations. These datasets provide language explanations, which can enhance the model's explainable reasoning capabilities.CoVLA-Dataset [23] employs automated methods to generate data with visual inputs, future trajectories, and detailed natural language descriptions. Impromptu VLA Dataset [27] focuses on unstructured scenarios.

Although existing datasets each have their strengths, there is currently no single dataset that can meet all needs. Future research needs to integrate the advantages of these datasets to build a comprehensive dataset that includes both multimodal sensor information and high-quality language annotations, in order to comprehensively enhance the perception, reasoning, and planning performance of VLA models.

IV. Challenges and Future Directions

Compared to data-driven end-to-end approaches, VLA models demonstrate advantages in multiple aspects. However, they still face several practical challenges.

A. Challenges

1) Latency and Deployment Challenges: large-scale models face high inference latency in on-vehicle deployment [28][29][30][31][33], constrained by heavy reliance on GPU computational power, substantial memory requirements, and autoregressive characteristics. For example, the average inference time of the AutoVLA [22] model on the nuPlan dataset is 1.3 seconds, while real-time in-vehicle deployment typically requires millisecond-level latency far below this.

2) Insufficient Robustness in Closed-Loop Environments: current evaluation frameworks primarily operate in open-loop [29], non-interactive [31], or simulated environments [33], lacking real-time environmental interaction. Open-loop evaluation mainly focuses on optimizing metrics such as L2 error based on datasets like nuScenes, which limits the system's robustness in interactive closed-loop real-world traffic, resulting in degraded performance when models are deployed in actual environments.

3) Hallucination Issues: large models in autonomous driving face significant hallucination challenges [52][53], manifesting as misjudgments in complex traffic scenarios and inconsistencies in processing multimodal data, potentially leading to erroneous driving decisions, compromising safety, and undermining real-time inference reliability.

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TABLE I
VLA MODELS IN AUTONOMOUS DRIVING

Action decoder	Model	Vision encoder	LLMs/VLMs	Dataset	Contribution
Autoreg-ressive Model	AutoVLA [22]	ViT	Qwen2.5-VL-3B	nuPlan,Waymo end-to-end driving,nuScenes,CARLA-Garage,DriveLM dataset	Unified reasoning, action generation, Adaptive reasoning via reinforcement
	CoVLA-agent [23]	CLIP ViT-L	LLaMA2-7B	CoVLA-Dataset	CoVLA: Large, comprehensive driving dataset, Automated captioning and trajectory generation
	EMMA [26]	Gemini-VE	Gemini	Waymo Open Motion Dataset,nuScenes,Waymo Open Dataset	End-to-end multimodal driving model Generalist model, multiple driving tasks
	Impromptu VLA [27]	ViT	Qwen2.5-VL-3B	Impromptu VLA	Open unstructured driving dataset, Improves VLA model performance
	OccLLaMA [28]	BEV encoder	LLaMA-3.1-8B	nuScenes,Occ3D, nuScenes-QA	Generative world model, Novel sparse scene tokenizer
	Opendrive VLA [29]	BEV encoder	Qwen2.5-Instruct	nuScenes,nuScenes-QA,nuCaption,nuX	End-to-end VLA model, Hierarchical vision-language alignment
	Sce2DriveX [32]	OpenCLIP	Vicuna 1.5 7B	nuScenes,Bench2Drive	Human-like CoT reasoning MLLM, Comprehensive 3D VQA dataset
	SimLingo [33]	InternViT-300M	Qwen2-0.5B	Bench2Drive, DriveLM dataset	Vision-only closed-loop driving, Language-action alignment
	VLA4CD [34]	SigLIP+DinoV2	LLaMA7b	CARLA	Simultaneous chat, action decisions, Continuous action values
	WiseAD [35]	CLIP ViT-L/14	MobileLLaMA-1.7B	LingoQA,DRAMA,BDDX, DriveLM,HAD,CARLA	Knowledge-augmented VLM for autonomous driving, Joint training improves driving performance
	DriveGPT-4 [36]	CLIP ViT-L/14	LLaMA2-7B	BDD-X	Interpretable end-to-end driving, New driving instruction dataset
	DriveGPT4-V2 [37]	MV-VT	Qwen-0.5B/TinyLLaMA	CARLA	Closed-loop autonomous driving, LLMs for online imitation learning
	RAG-Driver [38]	ViT-B/32	Vicuna 1.5 7B	BDD-X,nuScenes	Retrieval-augmented in-context learning, Generalisable driving explanations
	SafeAuto [39]	ViT-B/32	Vicuna 1.5 7B	BDD-X,DriveLM	Knowledge-enhanced safe autonomous driving, Novel PDCE loss for control
	Lingo-2 [40]	Wayve Vision Model	Wayve Vision Language Action Model	N/A	First public road VLAM
	Openemma [50]	CLIP、ViT	LLaVA-1.6-Mistral-7B/ Llama-3.2-11B-Vision-Instruct/ Qwen2-VL-7B-Instruct	nuScenes, nuImages dataset	Open-source MLLM framework, Enhanced end-to-end AD
	FSDrive [51]	ViT	Qwen2-VL-2B	nuScenes,DriveLM dataset	Spatio-temporal CoT, Unified pre-training paradigm
Diffusion Model	DiffVLA [24]	CLIP ViT-L/14	Vicuna 1.5 7B	NavSim,nuScenes, NAVSIMv1,nuPlan	Hybrid sparse-dense diffusion policy, VLM guidance for planning
	DriveMoE [25]	SigLIP-So400m	PaliGemma	Bench2Drive,CARLA	MoE for VLA driving, Enhances vision and action
	ORION [30]	EVA-02-L	Vicuna v1.5	Bench2Drive,nuScenes	Vision-language guided diffusion planning, Hybrid sparse-dense diffusion policy
	ReCogDrive [31]	InternViT	Qwen2.5-7B	NAVSIM,nuScenes	Reinforced VLM for AD, Integrates VLM with diffusion planner

4) Inadequate advanced reasoning capabilities: current models primarily rely on implicit [30] or basic reasoning mechanisms, with advanced methods like Chain-of-Thought [50] still underdeveloped. This limitation restricts their performance in unstructured or dynamic environments.

B. Future Directions

1) Optimizing real-time inference efficiency: model compression techniques such as quantization, distillation [54], and pruning, as well as efficient architecture design [55], such as Mixture of Experts [25], can significantly reduce computational overhead.

2) Enhancing spatial reasoning capabilities: by generating high-fidelity and diverse 3D driving scenario data, combined with spatial perception-enhanced pretraining, such as representation learning based on BEV or point clouds, the robustness and training efficiency of spatial reasoning can be improved. Meanwhile, on the vehicle side, by integrating large language models with spatial intelligence, such as Mind

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GPT [56], dynamic environment spatial modeling and rolling prediction can be achieved, thereby optimizing the precision and safety of closed-loop decision-making [8].

3) Developing world models: on one hand, world models generate diverse and high-fidelity driving scenario data [60] to support the training of VLA models, enhancing their robustness and data efficiency in complex environments. On the other hand, deploying world models on the edge enables real-time environment modeling and rolling prediction [61], integrating perception, prediction, and planning to achieve closed-loop decision-making.

4) Utilizing reinforcement learning: Reinforcement learning strategies such as Group Relative Policy Optimization [21][57][58] can effectively address the multimodal characteristics of planning and the issue of sparse rewards, thereby enhancing training stability and efficiency. Moreover, by utilizing a safe reinforcement learning framework like VL-SAFE [59] for interactive closed-loop simulation, the model's ability to interact and make real-time decisions in dynamic traffic environments can be trained, thus accelerating closed-loop simulation.

V. Conclusion

The VLA model, as an emerging paradigm in the field of autonomous driving, provides a unified framework by integrating visual perception, language reasoning, and action generation, aiming to enhance the system's interpretability and adaptability in complex environments. This paper presents a systematic review of the latest advancements of VLA models in autonomous driving, elucidating the necessity of VLA models in this domain, summarizing their typical architectures, reasoning processes, and related datasets, and discussing current challenges and future research directions.

Despite the multifaceted advantages of VLA models, they still face several practical challenges. Future research should focus on addressing these challenges, including reducing inference latency and deployment costs, improving robustness in closed-loop environments, addressing hallucination issues, and enhancing high-level reasoning capabilities. By tackling these challenges, the development and deployment of safer, more reliable, and more efficient VLA-based autonomous driving systems can be promoted.

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